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SPACE SCIENCE WITH PYTHON

Space Science with Python — Asteroid Project (Part 1)

Part 21 of the tutorial series starts with a first science project: The detectability of so called Near-Earth Objects (NEOs).

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Preface

This is the 21st part of my Python tutorial series "Space Science with Python". All codes that are shown here are uploaded on GitHub. Enjoy!

Motivation

In our last 20 *Space Science with Python* tutorials we have learned some astro-dynamical basics, miscellaneous Python libraries and tools that are helpful for a space scientist and we worked on a few use cases (like the comet 67P, the movement of Venus in the sky any many more).

We have a solid skillset to start working on a first science project. This science project shall cover miscellaneous tasks and ideas that I will address in the following.

Please note: we will not work on a scientific paper. Though some scientific insights could be merged and written in a publication. The purpose of this project is to get a feeling how one could define and work on a scientific objective. Further, by developing a Python library we provide a sustainable solution that can be used by us and other (free-time) scientists to extend the ideas.

1. Near-Earth Objects

It's the 15th February 2013. A bus-sized asteroid enters the atmosphere, bursts and explodes a few kilometres above the ground, causing a shock wave that damages property and injures several people in the Russian city Chelyabinsk.

Why did that happen? Well… our cosmic vicinity is populated with thousands or hundred-thousands of so called minor bodies; asteroids, comets and meteoroids. These objects are located in different regions of our Solar System, creating miscellaneous structures like:

- The Asteroid Main Belt
- The Kuiper Belt
- The Oort Cloud

… and sub-types and groups like the so called Trojans, Greeks or Hildas.

We do not need to dive deeper into this scientific topic. However, one particular "type" of class can be quite hazardous for our blue planet: **Near-Earth Objects**, short: **NEO**s.

NEOs are quite simple defined: All objects with a perihelion of equal / less than 1.3 AU. That's it.

Consequently, NEOs approach planet Earth quite closely, some cross our orbit and some, if not detected and catalogued before, may enter Earth's atmosphere. If the object is large enough, the resulting collision can be devastating, as in Chelyabinsk.

Space Agencies, astronomers, and so called all-sky surveys however monitor these objects. They scan the night sky for new, unknown NEO candidates enlarge our knowledge of these potentially hazardous objects with every observations night. Nasa …

Center for NEO Studies

NASA's Near-Earth Object (NEO) web-site. Data related to Earth impact risk, close-approaches, and much more.

cneos.jpl.nasa.gov

… as well as ESA …

ESA - European Space Agency

What is even more impressive is that the final dataset, collected over just two hours, is sufficient to determine the…

neo.ssa.esa.int

… provide nice public outreach information pages with all relevant information.

One webpage by ESA called NEODYS gathers all scientific NEO data, like orbital and physical parameters of all known NEOs. NEODYS is like the Minor Planet Center, but focusing exclusively on NEOs:

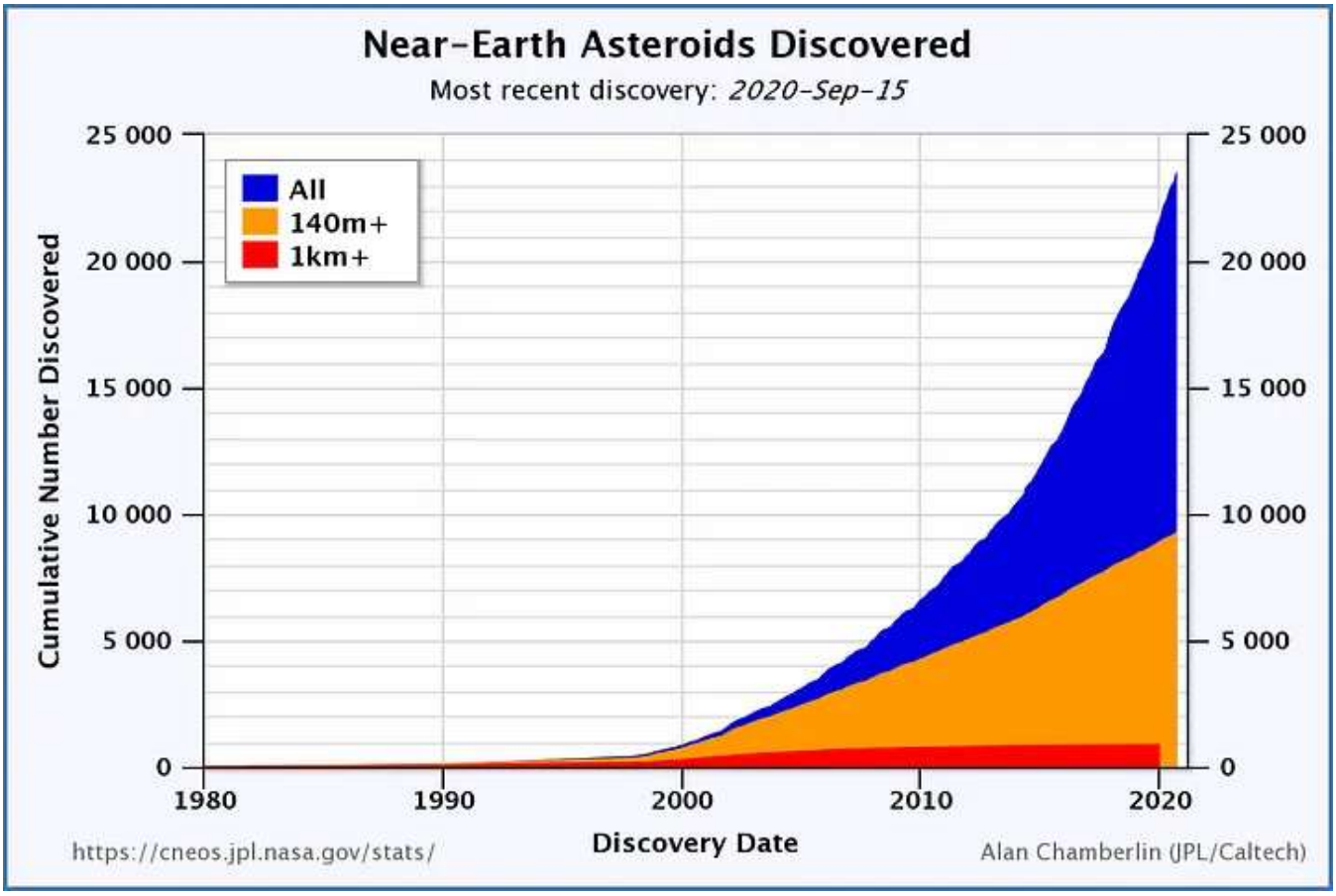
NEODyS

NEODyS provides information and services for all Near-Earth Asteroids (NEAs). Each NEA has its own dynamically...

newton.spacedys.com

Today (18.09.2020) there are 23,662 NEOs in the database.

Every day, more and more NEOs fill these databases. Currently, on average, we find 3 to 5 new objects every night. 8 years ago, when I wrote my Bachelor thesis about the detectability of NEOs, around 9000 objects were known. Now it is almost a factor of 3 more.



Cumulative number of NEO discoveries vs. the date. Three colours indicate NEO sizes. From: https://cneos.jpl.nasa.gov/stats/totals.html

The plot above shows the cumulative number of discoveries vs. the date, divided into 3 size-depending groups (diameter of 1 km and more, 140 m and more, and all). As you can see, the number of kilometre sized objects reached a plateau. Hopefully we found all “planet killers”... however objects with a size between 100 m and 1000 m are still being found on a regular basis. Thanks to modern telescopes and telescope surveys, observation techniques, data reduction pipelines and detection

algorithms we find more and more. The question is: ***how many more are left to discover?*** And since we need to operate telescopes and surveys to find all objects: ***What is the best "strategy" to detect all objects?"***.

Are these questions even "answerable"? Well, that's why we are here. The scientific questions stated above are rather important since they may help us, and observational astronomers to estimate the ongoing effort of detecting and cataloguing all objects.

Previous work has already been done in this regard and ESA also published a freely available toolkit to compute observational estimations called *Near-Earth Object Population Observation Program (NEOPOP)*.

ESA - European Space Agency

During activities related to ESA's Space-Situational Awareness (SSA) programme, the need for an observation system...

neo.ssa.esa.int

However, the task is challenging and a good first project to work on for the next couple of sessions.

Our observations of NEOs are probably biased, and using already known objects for this kind of work would be redundant. What we need is a state-of-the-art NEO population model that shall help us to estimate the detectability of the estimated over-all NEO population. After some literature research on the astrophysics data system (ADS) one finds quickly the most recent model by Granvik et al. 2018: Debiased orbit and absolute-magnitude distributions for near-Earth objects. The paper is freely accessible.

2. Library Development

The detectability of NEOs will be a complex topic with several pitfalls, statistical and numerical analysis and so on. Think about the different types of telescopes (size, camera systems, ground based — space based, etc.) that may effect the detectability of a Near-Earth Object. Instead of picking one telescope with certain pre-defined

parameters, we will chose a more generic approach: The development of a Python library that shall be useable by anyone, any astronomer and scientist to estimate the detectability of NEOs with their telescope/instrument specific settings!

So, our science project will be also a software engineering task as well! At the end, we can choose some telescope settings and run some simulations to derive some conclusions. At the end, we have a sustainable and extendable open source toolkit that fulfils also a reasonable purpose!

3. Test Driven Development

Developers experience a lot of different working setting and frameworks, like Waterfall or an Agile environment. Currently, a lot of frameworks, tools and planning methods are being used (or wrongly used) like KanBan, Scrum, SAFe, LeSS … you name it.

One approach emerged in the last couple of years that appears to be boring and tedious: Test-Driven Development (TDD). Usually, or let's say, most developers and free-time coders *work all over the place*: They want to see prototypes and results quickly, without any refactoring, commenting or coding. At the end of the development cycle, some do not implement even a unit test because "it just works". Maintaining, extending and understanding the resulting code can be exhausting and almost impossible for others (and even for the developer him- or herself if the developer does not work on the code frequently).

Our science project and development process will include a TDD approach to deliver a reliable, trusted and re-usable product. We will also fulfil PEP8 and PEP257 standards, follow Numpy Docstring conventions and will also use profiling to optimize the code. All in all the next sessions will cover:

- Space Science — Especially NEOs, Asteroids, etc.
- Numerical simulations; working with unbiased models
- Development of a Python library
- Test Driven Development

- PEP standards
- NumpyDoc
- Code optimization

I am looking forward to this project! It took me some time to evaluate a nice project for all of us — So let's tackle this task!

Thomas

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